The Riemann Hypothesis: Foundations, Obstructions, and Reorientation

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Abstract

This paper, the first in a five-part series on the Riemann Hypothesis, establishes the mathematical landscape and documents the systematic exploration—and ultimately the failure—of a gauge-theoretic approach. In the first part, we develop the analytical framework: the Riemann Hypothesis as a stability problem, the Li criterion as the central tool, and the Bombieri–Lagarias decomposition of the Li coefficients into archimedean and finite-part contributions. Starting from a thermodynamic formalism (Prime Hub Graphs and Ruelle transfer operators), we prove several unconditional structural results: the archimedean–finite decomposition, the archimedean dominance paradox, OS reflection positivity, the arithmetic uncertainty relation, and the spectral census of the arithmetic kernel.

In the second part, we identify four critical obstructions (K1–K4) that individually and collectively doom the original proof strategy: the Gibbs normalization gap, the sign change of the target functional, the wrong monotonicity direction, and the trace-class obstruction. We localize the difficulty precisely through a nine-step logical chain, showing that the *entire* content of RH concentrates in a single step (S5: the unconditional positivity of the Li coefficients). We formulate five concrete open problems and identify the reorientation toward Connes' spectral program as the path forward. Part II develops this new direction via the Shift Parity Lemma, finite-range diagnostics, and a conditional even-dominance reduction; Part III surveys the spectral routes; Part IV collects cross-route synthesis and remaining paths; Part V draws the final conclusions. In v2.0, two non-existence theorems in Part II (NE-A, NE-B) formally rule out the absolute total-positivity and universal-commuting-operator routes, reinforcing the arithmetic-statistical nature of even dominance.

Series status (v3.0, 2026-05-26). Following external critical review, the series status is revised from "unconditional A1–A8" (v2.1) to *conditional reduction*: even dominance is rigorously certified on the finite range $100 \leq \lambda \leq 1{,}300{,}000$ via separate IA-upper/lower bounds for $\lambda_1^{\pm}$; the asymptotic argument establishes only the variational statement $\lambda_{\min}(W^+ - W^-) = -\Theta(\sqrt{\lambda})$ and the eigenvalue inequality $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$ asymptotically remains conditional on the *Asymptotic Variational Gap Conjecture* for λ_1^- (Part II, "Status of the proof" remark, v2.2 revision; companion paper [10]). The Shift Parity Lemma, Leading-Mode Cancellation, the 33 finite-range diagnostics, and the non-existence theorems NE-A and NE-B are unaffected by this revision. The series is a continuously evolving draft/preprint; v3.0 restructures the trilogy into a five-part series (structural reorganization only; the conditional reduction status from v2.2 is unchanged).

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The Landscape

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. Reformulated via Li's criterion [1], RH is equivalent to the positivity of an infinite sequence of real numbers:

$$\text{RH} \iff \lambda_n \geq 0 \text{ for all } n \geq 1, \tag{1}$$

where $\lambda_n = \sum_\rho [1 - (1 - 1/\rho)^n]$ and the sum runs over the non-trivial zeros of ζ.

This paper is the first in a five-part series:

- **Part I** (this paper): Foundations, obstructions, and reorientation — mathematical foundations, analytical tools, structural results, the documentation of four dead ends, the localization of difficulty, and the reorientation toward Connes' spectral program.
- **Part II** [9]: Even dominance — the Shift Parity Lemma, computer-assisted finite-range diagnostics, and the conditional reduction architecture.
- **Part III** [12]: Spectral routes — Branch B (I-1 normal-family / err-collapse) and Branch Z (CCM microcluster programme), including spectral dead ends.
- **Part IV** [13]: Cross-route synthesis and remaining paths — inter-branch diagnostics, dormant and external routes, and methodological dead ends.
- **Part V** [14]: Conclusio — what is proven, what is excluded, and what remains.

The series documents an honest research journey: from a promising gauge-theoretic approach (Sections 2–6), through its failure at multiple points (Sections 9–12), to a new structural understanding of the problem via the Weil quadratic form (Section 16), and a final accounting of what has been achieved (Part V).

1.1 The Hope

The Hilbert–Pólya conjecture proposes that the non-trivial zeros correspond to eigenvalues of a self-adjoint operator. The Berry–Keating semiclassical approach ($\hat{H} = xp$) fails due to the Weyl law obstruction. This paper shifts the paradigm from quantum mechanics to statistical mechanics: the zeros are generated not as eigenvalues of a Hamiltonian, but as resonances of a Ruelle-type transfer operator.

The guiding idea is that the Riemann Hypothesis is a *stability condition*: the arithmetic structure of the primes, encoded in the Li coefficients λ_n, is stabilized by a competition between arithmetic instability (from the primes) and archimedean convexity (from the Gamma function). Understanding this competition is the central theme of the series.

2 The Thermodynamic Formalism

2.1 Prime Hub Graph and the Vacuum Trace

To reproduce the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, we construct a dynamical system whose primitive periodic orbits correspond to primes.

Definition 2.1 (Prime Orbit Axiom)**.** We postulate a topologically mixing directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ whose primitive cycles γ_p are in bijection with the primes $p \in \mathcal{P}$, with geometric length $\ell(\gamma_p) = \log p$.

Remark 2.2 (Motivation vs. logical dependency). The Prime Orbit Axiom is a constructive motivation. The analytical results in subsequent sections depend solely on properties of $\zeta(s)$ and the arithmetic Euler product, and are logically independent of this graph construction.

Using a holonomy filter on the free monoid generated by $\mathcal{P}$, we define a transfer operator $\mathcal{L}_s$ and prove:

Theorem 2.3 (Recovery of the Zeta Function)**.** *For $\Re(s) > 1$, the Fredholm determinant of the vacuum-projected transfer operator satisfies:*

$$\det(I - \mathcal{L}_s)^{\text{vac}} = \prod_p (1 - p^{-s}) = \frac{1}{\zeta(s)}.$$

Proof. By the Ruelle identity [3], $\log \det(I - \mathcal{L}_s) = -\sum_{m=1}^{\infty} \frac{1}{m} \operatorname{Tr}(\mathcal{L}_s^m)$. The vacuum projection restricts the trace to configurations that form closed loops of prime powers: the only periodic orbits of length $m \log p$ in $\mathcal{G}$ are the m-fold traversals of the primitive cycle γ_p, each contributing weight p^{-ms} with multiplicity $\log p$ (the geometric length). Summing over prime powers and exponentiating:

$$\log \det(I - \mathcal{L}_s)^{\text{vac}} = -\sum_p \sum_{m=1}^{\infty} \frac{1}{m} p^{-ms} = \sum_p \log(1 - p^{-s}).$$

This is the logarithm of the Euler product $\prod_p (1 - p^{-s}) = 1/\zeta(s)$, valid for $\Re(s) > 1$ by absolute convergence. □

2.2 The MEPP Heuristic

The Maximum Entropy Production Principle (MEPP) interprets the critical line $\Re(s) = 1/2$ as a detailed-balance condition:

Heuristic Principle 2.4 (Model RH via MEPP). If the primes represent the optimally stable configuration of a "prime gas" subject to the PNT density constraint, the resulting Gibbs measure satisfies time-reversal symmetry, forcing zeros onto the critical line.

This heuristic does not constitute a proof but motivates the thermodynamic perspective developed below.

3 The Li Coefficients: Analytical Framework

3.1 Bombieri–Lagarias Representation

The completed xi function $\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ provides the definitive representation:

Theorem 3.1 (Bombieri–Lagarias [2])**.** *For $n \geq 1$:*

$$\lambda_n = \frac{1}{(n-1)!} \frac{\partial^n}{\partial s^n} \left[s^{n-1} \log \xi(s) \right]_{s=1}.$$

Since $\xi(s)$ is entire of order 1 with $\xi(1) = 1/2 \neq 0$, $\log \xi(s)$ is holomorphic near $s = 1$ and the derivative is well-defined.

3.2 Archimedean–Finite Decomposition

Proposition 3.2 (Archimedean–Finite Decomposition)**.** *Each Li coefficient decomposes as* $\lambda_n = \lambda_n^{(\infty)} + \lambda_n^{(\mathrm{fin})}$*, where:*

$$\lambda_n^{(\infty)} = \frac{1}{(n-1)!}\frac{\partial^n}{\partial s^n}\left[s^{n-1}\log\bigl(\tfrac{1}{2}s\,\pi^{-s/2}\,\Gamma(s/2)\bigr)\right]_{s=1}, \tag{2}$$

$$\lambda_n^{(\mathrm{fin})} = \frac{1}{(n-1)!}\frac{\partial^n}{\partial s^n}\left[s^{n-1}\log g(s)\right]_{s=1}, \quad g(s) := (s-1)\zeta(s). \tag{3}$$

Both parts are individually computable; $g(s)$ *is holomorphic at* $s=1$ *with* $g(1)=1$*.*

Remark 3.3 (Three-regime structure)*.* Numerical computation reveals:

(a) **Small** n **(**$n \lesssim 8$**):** $\lambda_n^{(\infty)} < 0$, compensated by $\lambda_n^{(\mathrm{fin})} > 0$. Positivity is a delicate cancellation.

(b) **Transition (**$n \approx 8$**):** $\lambda_n^{(\infty)}$ crosses zero.

(c) **Large** n **(**$n \gg 10$**):** $\lambda_n^{(\infty)} \sim \frac{n}{2}\log n$ dominates; positivity is automatic.

The "hard" instances of Li's criterion concentrate in regime (a). Table 1 shows the decomposition for $n = 1,\dots,13$.

n	λ_n	$\lambda_n^{(\infty)}$	$\lambda_n^{(\mathrm{fin})}$	$c_n = \lambda_n/n$
1	0.023	-0.554	0.577	0.023
2	0.092	-0.875	0.967	0.046
3	0.208	-1.013	1.221	0.069
5	0.576	-0.883	1.458	0.115
8	1.466	$+0.021$	1.445	0.183
10	2.279	$+0.956$	1.324	0.228
13	3.817	$+2.725$	1.093	0.294

Table 1: Archimedean–finite decomposition of the Li coefficients. The sequence $c_n = \lambda_n/n$ is strictly increasing, ruling out complete monotonicity.

4 The Archimedean Dominance Paradox

Proposition 4.1 (Stability Decomposition)**.** *The global curvature* $\mathcal{M}(\sigma) = \partial_s^2 \log\xi(s)$ *decomposes as* $\mathcal{M} = \mathcal{M}_{\mathrm{arith}} + \mathcal{M}_{\mathrm{arch}}$*:*

1. ***Arithmetic instability:*** $\mathcal{M}_{\mathrm{arith}}(1) = \sum_\rho \Re(1/\rho^2) < 0$*.*

2. ***Archimedean stability:*** $\mathcal{M}_{\mathrm{arch}}(\sigma) > 0$ *for* $\sigma > 1/2$ *(explicitly positive and convex).*

The Riemann Hypothesis is equivalent to the archimedean convexity globally dominating the arithmetic concavity for all $\sigma > 1/2$. This "dominance paradox" — a manifestly positive continuous term must overcome infinitely many oscillatory arithmetic contributions — is the structural core of the problem.

5 Structural Results

We establish five unconditional results that constrain the problem and form the toolkit for subsequent sections and papers.

5.1 Prime-Sum Functionals and the Gibbs Framework

Definition 5.1 (Arithmetic Gibbs Measure)**.** The Gibbs measure is $\mu_\sigma(p) = p^{-\sigma}/P(\sigma)$ with $P(\sigma) = \sum_p p^{-\sigma}$. The prime-sum functionals are:

$$A(\sigma) := \sum_p \frac{\log p}{p^\sigma(p^\sigma - 1)}, \qquad B(\sigma) := \sum_p \frac{\log p}{p^\sigma},$$

$$C(\sigma) := \sum_p \frac{(\log p)^2}{p^\sigma(p^\sigma - 1)}, \qquad D(\sigma) := \sum_p \frac{(\log p)^2}{(p^\sigma - 1)^2}.$$

Lemma 5.2 (Finite-Part Convergence)**.** *$A(\sigma)$, $C(\sigma)$, and $D(\sigma)$ converge absolutely at $\sigma = 1$ and are right-continuous. $B(\sigma)$ and $P(\sigma)$ diverge as $\sim \log(1/(\sigma - 1))$.*

Remark 5.3 (Gibbs normalization gap). The Gibbs-normalized expectation $-A(\sigma)/P(\sigma)$ loses the finite-part information as $\sigma \to 1^+$: the $1/P(\sigma)$ normalization annihilates $\lambda_n^{(\mathrm{fin})}$. This is the first hint that the thermodynamic approach has structural limitations (see Section 9).

5.2 Non-Identification Theorem

Proposition 5.4 (Non-Identification)**.** *The excess free energy $I_n := \lim_{\sigma\to 1^+} \Phi_n(\sigma)$ and the Li coefficient λ_n are generically* not *equal: $I_n \neq \lambda_n$. The former is a KL divergence (tilt cost), the latter a spectral trace of $\log \xi$.*

5.3 Variance Lower Bound

Proposition 5.5 (Variance Bound)**.** *$I_1 \geq \frac{1}{2}V_1$ where $V_1 = \lim_{\sigma\to 1^+} \mathrm{Var}_{\mu_\sigma}(H_{1,\sigma}) > 0$. Numerically, $I_1 \geq 0.034 > \lambda_1 \approx 0.023$.*

5.4 OS Reflection Positivity

Proposition 5.6 (OS Reflection Positivity)**.** *The Osterwalder–Schrader form*

$$K(t,u) = \sum_{p\in\mathcal{P}} w_p\, e^{-i(u-t)\log p}, \qquad w_p = \frac{p^{-(\sigma+1)}}{P(\sigma)} > 0,$$

is positive definite on $L^2_+(\mathbb{R})$. The OS Hilbert space carries a contractive semigroup $e^{-\tau H}$ with $H = \mathrm{diag}(\log p)$.

Proof. By Bochner's theorem: $K(t,u) = \hat{\mu}(u-t)$ with $\mu = \sum_p w_p\, \delta_{\log p}$ a positive measure, so $\iint \bar{f}(t) f(u) K(t,u)\, dt\, du = \sum_p w_p |\hat{f}(\log p)|^2 \geq 0$. $\square$

5.5 Arithmetic Uncertainty Relation

Heuristic Principle 5.7 (Arithmetic Uncertainty Relation). For normalized $\psi \in \ell^2(\mathcal{P})$, define the prime-domain spread Δ_D via

$$\Delta_D^2 = \sum_p |a_p|^2 (\log p - \overline{\log p})^2$$

and the zero-domain spread Δ_t via the Fourier transform $\hat{\psi}(t) = \sum_p a_p e^{-it\log p}$. Then $\Delta_D \cdot \Delta_t \geq 1/2$.

Remark 5.8. This follows formally from the substitution $x_p = \log p$ and the Heisenberg–Kennard–Weyl inequality on $L^2(\mathbb{R})$. However, ψ is supported on the discrete set $\{\log p : p \in \mathcal{P}\}$, not on all of $\mathbb{R}$, so the reduction requires an embedding into $L^2(\mathbb{R})$ whose properties depend on the distribution of primes. We state this as a heuristic principle, not a theorem.

The physical content is a consistency check: a zero off the critical line would require a "localized phase conspiracy" among finitely many primes, which the arithmetic incommensurability of $\{\log p\}$ makes implausible but does not rigorously exclude.

5.6 Strict Convexity and the Spectral Census

Proposition 5.9 (Strict Convexity). *$F(\sigma) := -\log\zeta(\sigma)$ is strictly convex on $(1,\infty)$: $F''(\sigma) = \sum_p (\log p)^2 p^{-\sigma}/(1-p^{-\sigma})^2 > 0$.*

Proposition 5.10 (Spectral Census). *For the symmetrized arithmetic kernel K_σ^{sym} truncated to $N \geq 300$ primes and $\sigma \in (1,5)$, the negative inertia index is exactly 2, independent of N and σ.*

6 The Gauge Equivalence Framework

The spectral census (Theorem 5.10) suggests a gauge-correction approach: can a rank-2 correction restore positive-semidefiniteness?

Definition 6.1 (Admissible Gauge). A symmetric kernel G_σ on $\mathcal{P} \times \mathcal{P}$ is admissible if $\sum_{p,q} G_\sigma(p,q)\mu_\sigma(p)\mu_\sigma(q) = 0$.

Theorem 6.2 (Gauge Equivalence). *For fixed $\sigma > 1$, the following are equivalent:*

(i) The target functional $F(\sigma) = A(\sigma)B(\sigma) - P(\sigma)[C(\sigma) + D(\sigma)] \geq 0$.

(ii) There exists an admissible gauge making $K_\sigma^{\mathrm{sym}} + G_\sigma \succeq 0$.

This tautological equivalence provides the formal structure for the gauge-theoretic approach. Whether it leads anywhere depends on the *global* behavior of $F(\sigma)$ — a question addressed in Section 10, where we discover that $F(\sigma) < 0$ for $\sigma > 1.14$.

7 Summary: The Starting Position

At the end of the landscape analysis, we have assembled the following toolkit:

Result	**Status**
R1: Bombieri–Lagarias representation	proven
R2: Archimedean–finite decomposition	proven
R3: Non-identification ($I_n \neq \lambda_n$)	proven
R4: Variance lower bound ($I_1 \geq \frac{1}{2}V_1$)	proven
R5: Arithmetic uncertainty relation	heuristic
R6: Strict convexity of $-\log\zeta$	proven
R7: OS reflection positivity	proven
R8: Spectral census (index = 2)	numerical
R9: Gauge equivalence (tautological)	proven

The hope is clear: use the gauge framework (R9) together with the spectral census (R8) to construct a global certificate proving $\lambda_n \geq 0$. The following sections will show that this hope is dashed by a sign change at $\sigma_0 \approx 1.14$ — but that a completely different approach, through Connes' Weil quadratic form, opens a new and more promising path.

The Dead Ends

8 The Original Strategy

The toolkit of nine structural results (R1–R9) and the gauge equivalence framework established above suggest a clear proof strategy:

1. Use the spectral census (R8: negative inertia index = 2) to identify the unstable modes.
2. Construct an explicit rank-2 gauge correction restoring positive-semidefiniteness.
3. Apply the gauge equivalence theorem (R9) to deduce $\lambda_n \geq 0$.

This strategy fails at step 2. In this part, we document four independent obstructions that kill it, then use the resulting understanding to localize the difficulty precisely.

9 Dead End 1: The Gibbs Normalization Gap (K1)

The thermodynamic formulation defines the renormalized constraint $m_n(\sigma) = \mathbb{E}_{\mu_\sigma}[X_{n,\sigma}] - h_n(\sigma)$, with the hope that $\lim_{\sigma\to 1^+} m_n(\sigma) = \lambda_n$.

Proposition 9.1 (K1: Critical Identification Fails)**.** *The Gibbs-normalized expectation $m_1(\sigma) = -A(\sigma)/P(\sigma) - h_1(\sigma)$ does* not *converge to λ_1 as $\sigma \to 1^+$. The $1/P(\sigma)$-normalization annihilates the finite-part contribution $\lambda_1^{(\mathrm{fin})} = \gamma \approx 0.577$.*

Proof. The numerator sum $S(\sigma) = \sum_p (\log p)p^{-\sigma} f(p, \sigma)$ converges to a finite limit, while $P(\sigma) \to \infty$. Therefore $-S(\sigma)/P(\sigma) \to 0$. The Li coefficient λ_1 arises from $\frac{d}{ds} \log \xi(s)|_{s=1}$, which involves $\zeta'(s)/\zeta(s) + 1/(s-1) \to \gamma$ — a cancellation that occurs *before* Gibbs normalization, not after. □

Remark 9.2 (Consequence). The excess free energy I_n and the Li coefficient λ_n are structurally distinct objects (Theorem 5.4). Even $I_n \geq 0$ (which holds unconditionally as a KL divergence) does not imply $\lambda_n \geq 0$. This is the "convexity trap": by the Bregman inequality, $I_n \geq 0$ is a universal identity for any strictly convex generating function, independent of $\operatorname{sign}(\lambda_n)$.

10 Dead End 2: Global Gauge Positivity Fails (K2)

The gauge equivalence theorem (R9) requires $F(\sigma) \geq 0$ for all $\sigma > 1$.

Proposition 10.1 (K2: Sign Change of $F(\sigma)$). *The target functional $F(\sigma) = A(\sigma)B(\sigma) - P(\sigma)[C(\sigma) + D(\sigma)]$ satisfies:*

- *$F(\sigma) > 0$ for $\sigma \in (1, \sigma_0)$ with $\sigma_0 \approx 1.14$,*
- *$F(\sigma) < 0$ for all tested $\sigma > \sigma_0$ (minimum $F \approx -0.184$ at $\sigma \approx 1.28$).*

This sign change is independent of RH.

Proof. Near $\sigma = 1 + \varepsilon$: $A \cdot B \sim 1/\varepsilon$ dominates $P(C + D) \sim \log(1/\varepsilon)$, so $F > 0$. For larger σ: $B(\sigma)$ decays faster than $P(\sigma)$, and the product AB falls below $P(C + D)$. Numerical computation with $N = 10000$ primes and Rosser–Schoenfeld tail bounds confirms the transition at $\sigma_0 \approx 1.14$. □

Remark 10.2 (Consequence). The claim "RH $\Leftrightarrow$ $F(\sigma) \geq 0$ for all $\sigma > 1$" is **false**. The gauge approach works only in the boundary layer $(1, 1.14)$, not globally. No admissible gauge can make K_σ^{sym} positive-semidefinite for $\sigma > 1.14$.

11 Dead End 3: Monotonicity Direction Wrong (K3)

The original argument assumed that $\frac{d}{ds} \log \xi(\sigma)$ is decreasing, making λ_1 the maximum.

Proposition 11.1 (K3: Convexity, Not Concavity). *$\frac{d^2}{ds^2} \log \xi(\sigma) \approx +0.046 > 0$ at $\sigma = 1$. The function $\log \xi$ is* convex *near $s = 1$, not concave. Therefore λ_1 is the* minimum *of the derivative profile, not the maximum.*

Remark 11.2 (Consequence). This reverses the logic of the monotonicity argument: instead of showing $\lambda_1 \geq 0$ by bounding the maximum from below, one would need to bound the minimum from below — a strictly harder problem.

12 Dead End 4: The Stieltjes Realization and A8 (K4)

12.1 Path D2: Stieltjes Realization

Proposition 12.1 (K4a: Complete Monotonicity Fails). *The sequence $c_n = \lambda_n/n$ is* strictly increasing*: $c_1 \approx 0.023$, $c_{13} \approx 0.294$, with $c_n \sim \frac{1}{2}\log n$ asymptotically. This violates complete monotonicity at $k = 1$ ($\Delta c_n > 0$, so $(-1)^1 \Delta c_n < 0$). The Stieltjes realization approach with a measure on* $[0, 1]$ *is ruled out.*

Under RH, the spectral measure lives on the unit circle $|w_\rho| = 1$, not $[0, 1]$. The positivity of c_n is an essentially number-theoretic property.

12.2 K4b: Inconsistency of the Original A8

Proposition 12.2 (Hardy Contradiction). *The original axiom A8 posits $\mathcal{K}_0 \geq 0$ trace-class with $\xi(1/2+it) = C \cdot \det(I + V_t\mathcal{K}_0V_t^*)$. But $\mathcal{K}_0 \geq 0$ implies $\det(I + V_t\mathcal{K}_0V_t^*) \geq 1 > 0$ for all real t. This contradicts Hardy's theorem (1914): ξ has infinitely many zeros on the critical line.*

12.3 The Trace-Class Obstruction

The corrected axiom A8′ requires $\xi(s)/\xi(0) = \det(I - \mathcal{L}_s)$ with $\mathcal{L}_s$ nuclear. However, the Euler product gives $\mathcal{L}_s = \mathrm{diag}(p^{-s})$ on $\ell^2(\mathcal{P})$, which is trace-class only for $\Re(s) > 1$. At the critical line, $\sum_p p^{-1/2} = +\infty$.

For the Selberg zeta function, the analogous construction works because geodesic lengths grow exponentially ($\#\{\gamma : l_\gamma \leq T\} \sim e^T/T$), while prime "lengths" $\log p$ are too dense ($\pi(x) \sim x/\log x$). This density gap is the fundamental obstruction.

13 Route Analysis: What Survives

The axiom system (defined in Part II) admits two logical routes to RH:

Route 1 (A7 + A8′): A7 (OS reflection positivity) is **proven**. A8′ (nuclear Fredholm determinant for ξ) remains **open** and faces the trace-class obstruction.

Route 2 (A9): Dead. A9(i–ii) hold, but A9(iii) is broken by K1 and the convexity trap.

14 The Localization Table

We trace the logical chain from definitions to RH, identifying exactly where the difficulty concentrates.

Step	Statement	Status	Method
S1	Define $\xi(s)$	proven	definition
S2	λ_n via Bombieri–Lagarias	proven	[2]
S3	$\lambda_n = \lambda_n^{(\infty)} + \lambda_n^{(\mathrm{fin})}$	proven	Section 3
S4	$\lambda_n^{(\infty)} \sim \frac{n}{2} \log n$ for large n	proven	Stirling
S5	$\lambda_n \geq 0$ **for all** $n \geq 1$	**OPEN**	$\Longleftrightarrow$ **RH**
S6	All zeros on $\Re(s) = 1/2$	$\Leftrightarrow$ S5	Li (1997)
S7	$\pi(x) = \mathrm{Li}(x) + O(\sqrt{x} \log x)$	$\Leftarrow$ S6	von Koch
S8	Spectral census (index = 2)	numerical	Section 5
S9	Gauge equivalence (tautological)	proven	Section 6

Conclusion: The *entire* content of the Riemann Hypothesis concentrates in Step S5. Everything before it is unconditional; everything after it is either equivalent or a consequence. The present program has not reduced S5 to a simpler statement.

15 Five Open Problems

Based on our analysis, we formulate five concrete open problems ordered by estimated difficulty:

Conjecture 15.1 (OP1: Analytical Proof of Index Rigidity)**.** *The negative inertia index of K_σ^{sym} is exactly 2 for all N sufficiently large and $\sigma \in (1, \sigma_{\max})$. A proof should exploit the rank-2 hub structure of the transfer operator.*

Conjecture 15.2 (OP2: Uniform Bound on $\lambda_n^{(\mathrm{fin})}$)**.** $\lambda_n^{(\mathrm{fin})} > 0$ *for all* $n \geq 1$*. This is supported by the numerical observation that* $\lambda_n^{(\mathrm{fin})}$ *is always positive with slow algebraic decay.*

Conjecture 15.3 (OP3: Growth Rate)**.** $\lambda_n^{(\infty)} \sim \frac{n}{2} \log n$ *with explicit error bounds, uniform in n. This would settle RH for all n above an explicit threshold.*

Conjecture 15.4 (OP4: Small-n Positivity)**.** $\lambda_n > 0$ *for $n = 1, \ldots, n_0$ (some explicit n_0) unconditionally — without assuming RH. The first 10^4 coefficients are known to be positive [4].*

Conjecture 15.5 (OP5: Nuclear Transfer Operator for ξ)**.** *Construct a separable Hilbert space $\mathcal{V}$ and a nuclear family $s \mapsto \mathcal{L}_s$ with $\xi(s)/\xi(0) = \det(I - \mathcal{L}_s)$, spectral radius < 1 for $\Re(s) > 1/2$, and eigenvalue 1 at the zeros.*

OP5 is essentially the Hilbert–Pólya program and connects to Deninger's foliated spaces [8] and Connes' adelic approach [5].

16 Reorientation: Connes' Spectral Program

The failure of the gauge-theoretic approach (K1–K4) forces a fundamental reorientation. A recent survey by Connes [6] provides a new path: the Weil quadratic form QW_λ on $L^2([-\log \lambda, \log \lambda])$, constructed from the Weil explicit formula, defines a self-adjoint operator A_λ whose spectral properties are directly linked to the zeros of ζ.

Theorem 16.1 (Connes, Theorem 6.1 in [6]; proved in [7]). *If the minimum eigenvalue of A_λ is simple with even eigenfunction, then the Fourier transform of the eigenfunction has all zeros on the real line.*

Connes' Section 6.6 identifies the verification of this *even dominance* property as the remaining open step. Crucially, Connes' Hurwitz argument requires even dominance only along a sequence $\lambda_n \to \infty$, not for all λ.

This is where the thermodynamic insight pays off in an unexpected way: the decomposition analysis from the landscape sections (the prime-sum functionals, the trigonometric structure of shift operators) becomes the key to understanding the even-dominance route. Part II establishes the Shift Parity Lemma, the frontier-prime mechanism, and finite-range computer-assisted gap diagnostics for 33 values of λ up to 1,300,000, now treated as a conditional finite bridge until the odd-sector lower bound is independently validated.

Remark 16.2 (From despair to hope). The gauge-theoretic approach failed because it tried to prove a *global* statement ($\lambda_n \geq 0$ for all n) via a *local* tool (the Gibbs measure near $\sigma = 1$). The Connes approach succeeds where the gauge approach fails because it works directly in the spectral domain of A_λ, where the even-odd decomposition provides a natural structure. The Shift Parity Lemma (Part II) shows that this structure is fundamentally algebraic, not analytic.

Three Independent Branches Attack RH from the Same Reorientation

The spectral reorientation of Connes' programme opens *three* structurally independent attacks on RH, all of which appear in the present series. They share the same external preprint inputs (Connes 2026 [6], Connes–vS 2025 [7], and the Connes–Consani–Moscovici 2025 zeta-spectral-triples construction) and meet at the same analytical wall — Connes' §6.6(b)/MS2 condition on the limit behaviour of $\{\hat{\xi}_\lambda\}$ — but they reach the wall through different mechanisms.

- **Branch A — Even Dominance.** Reduces RH to the eigenvalue inequality $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$ along a sequence $\lambda_n \to \infty$. Developed in Part II of this series; standalone extraction in the companion paper [10]. Currently a conditional reduction (open: A3 odd-sector lower bound, A7 Asymptotic Variational Gap Conjecture).
- **Branch B — I-1 Normal-Family / err-Collapse.** Reduces RH to a single uniform growth-adapted strip bound (ii-a) on $\{\hat{\xi}_\lambda\}$ from QW_N-coercivity. Developed in Part III of this series, with reduction chain B1–B11, Foundation-Lock $\hat{\xi}(z) = (2/\sqrt{L})\sin(zL/2)F(z)$, and converged Mac Studio empirics ($A_\lambda(0.4) \approx 1.004$ flat across $\lambda \in \{3, 5, 7, 9, 11, 13, 15\}$).
- **Branch Z — CCM Microcluster Programme.** Reduces RH (via the err-collapse coupling and MS2-closure) to three named conditional inputs C2ca.1, C2ca.2, C2ca.5 plus a fixed-waterline outer-gap assumption $g_* \geq g_0$. Developed in the standalone companion paper [11]; summarised in Part III.

The three branches are pursued in parallel because each remaining open block sits at a structurally different place: A7 is variational, (ii-a) is analytical (strip bound from coercivity), C2ca is operator-theoretic (quasimode-to-projector). A success on any one of the three would suffice. The audit-trail philosophy of this paper preserves all three with explicit status, including a catalogue of dead routes already eliminated within each branch (Parts III–IV).

17 Conclusion

This paper has documented both a toolkit and an honest failure. The gauge-theoretic approach to the Riemann Hypothesis produced genuine structural insights (R1–R9) but cannot bridge the gap between the local thermodynamic analysis and the global spectral positivity required by Li's criterion. Four independent obstructions (K1–K4) kill the original strategy, and the localization table shows that no reduction to a simpler problem has been achieved: the difficulty remains concentrated in Step S5.

However, the failure is productive:

1. The landscape analysis (Sections 2–6) establishes a permanent toolkit of unconditional structural results.

2. The dead ends (K1–K4) are clearly mapped, preventing others from pursuing them.

3. The reorientation to Connes' program leverages the thermodynamic insight (prime shift operators, trigonometric structure) in a new context.

4. The understanding that even dominance is driven by prime contributions, not archimedean effects, emerged directly from the decomposition analysis.

Part II takes up the new thread: the Shift Parity Lemma, the computer-assisted finite-range diagnostics, and the identification of the cumulative eigenvalue-ordering step as the remaining challenge. Parts III–IV document the spectral routes, cross-route synthesis, and remaining paths; Part V draws the final conclusions on what has been proven, what has been excluded, and what remains open.

AI Disclosure

This manuscript was developed in extensive collaboration with AI language models (Claude, Anthropic; Copilot, Microsoft/GitHub; Gemini, Google DeepMind). AI contributed to conceptual exploration, analytical work, literature review, writing, and critical review. The author, Lukas Geiger, conceived the research direction, provided domain expertise, and exercised final editorial authority. The author assumes full and sole scholarly responsibility for all content, including any errors.

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